

Probability and Computing

Michaelmas Term 2023

Exercise Sheet 5

1. Give an FPRAS for the variant of DNF counting in which we are given a DNF formula and an integer k and we want to estimate the number of satisfying assignments in which exactly k of the variables are true.
2. Let $S_1, \dots, S_t$ be sets of indices in the range $\{1, \dots, n\}$. Say that a permutation π of $\{1, \dots, n\}$ is increasing on S_j if $\pi(i_1) < \pi(i_2) < \dots < \pi(i_{k_j})$ where $i_1 < i_2 < \dots < i_{k_j}$ are the elements of S_j .

Give an FPRAS that takes as input n and sets $S_1, \dots, S_t$, and approximates the number of permutations that are increasing on at least one of the S_j .

3. (a) Suppose, for some positive integer m , that we have m values $x_1, \dots, x_m$, each in the range $\frac{1}{3} < x_i < 1$. Let $x = \prod_{i=1}^m x_i$.
For each $i \in \{1, \dots, m\}$ we have an algorithm A_i that takes as input a rational error tolerance $\varepsilon \in (0, 1)$ and a failure probability $\delta \in (0, 1)$ so that its output $A_i(\varepsilon, \delta)$ satisfies $\mathbb{P}(1 - \varepsilon \leq \frac{A_i(\varepsilon, \delta)}{x_i} \leq 1 + \varepsilon) \geq 1 - \delta$. We wish to write an algorithm A' that takes input a rational error tolerance $\varepsilon' \in (0, 1)$ and a failure probability $\delta' \in (0, 1)$ and computes an estimate $A'(\varepsilon', \delta')$ such that $\mathbb{P}(1 - \varepsilon' \leq \frac{A'(\varepsilon', \delta')}{x} \leq 1 + \varepsilon') \geq 1 - \delta'$. The new algorithm A' may make calls to algorithms $A_1, \dots, A_m$. Give an algorithm A' that succeeds at this, and prove that it does.
- (b) Consider $i \in \{1, \dots, m\}$. Now we investigate how to produce algorithm A_i . Suppose that we have some set U_i of objects, and that x_i is the fraction of them that are in a subset H_i . Suppose that we have an algorithm B_i that samples uniformly from U_i . Using B_i , give an algorithm A_i that takes a rational error tolerance $\varepsilon \in (0, 1)$ and a failure probability $\delta \in (0, 1)$ so that its output $A_i(\varepsilon, \delta)$ satisfies $\mathbb{P}(1 - \varepsilon \leq \frac{A_i(\varepsilon, \delta)}{x_i} \leq 1 + \varepsilon) \geq 1 - \delta$. The number of times that A_i calls the uniform sampler B_i should be $O(\log(1/\delta)/\varepsilon^2)$.
- (c) Now we put Parts (a) and (b) together. In order to solve this part, you may need to re-consider your solution to Part (a). Suppose for each $i \in \{1, \dots, m\}$, that we have some set U_i of objects, and that x_i is the fraction of them that are in a subset H_i where $\frac{1}{3} < x_i < 1$. Suppose that B_i is an algorithm that samples uniformly from U_i . Let $x = \prod_{i=1}^m x_i$. Show how to write an algorithm A' that takes input a rational error tolerance $\varepsilon' \in (0, 1)$ and a failure probability $\delta' \in (0, 1)$ and computes an estimate $A'(\varepsilon', \delta')$ such that $\mathbb{P}(1 - \varepsilon' \leq A'(\varepsilon', \delta')/x \leq 1 + \varepsilon') \geq 1 - \delta'$. Show that the number of calls that A' makes to each B_i is $O(m^2(\log(m/\delta')/(\varepsilon')^2))$.
4. Let Δ and q be positive integers with $q \geq \Delta + 3$. A proper q -colouring of an n -vertex graph $G = (V, E)$ is an assignment $\sigma: V \rightarrow \{1, \dots, q\}$ such that every edge $(u, v) \in E$ has $\sigma(u) \neq \sigma(v)$. Show that given a PAUS for sampling the proper q -colourings of an input graph with maximum degree at most Δ , we can construct an FPRAS for approximately counting these colourings.

Hint: You can refer to calculation in the lecture notes where it is exactly the same — there is no need to copy it, but make sure to explain how you use the calculation there.

5. Consider the following Markov chain for shuffling n cards. In each step, we pick a card uniformly at random, out of the n cards. Then we pick another card independently and uniformly at random, out of the n cards. (This could be the same as the first card.) We exchange the positions of the two cards (if they are different).

Prove that, when it is started from an arbitrary permutation of the cards, this Markov chain will converge to the uniform distribution on permutations.

6. We repeatedly toss a fair coin and we use a counter to keep track of the length of the last run of heads. The counter has m bits and when it reaches $n = 2^m - 1$ it stays there as long as heads continue appearing. For example, when $m = 2$, for the sequence THHTHTTHHHHT $\dots$, the counter is in states $0, 0, 1, 2, 0, 1, 0, 0, 1, 2, 3, 3, 0, \dots$. This defines a Markov chain with state space $\Omega = \{0, \dots, n\}$ where the current state of the chain is the minimum of n and the length of the last run of heads.

- (a) For $m = 2$, write down the transition matrix P of this Markov chain.
- (b) What is the stationary distribution π of the Markov chain? (In general, not just in the $m = 2$ case.)
- (c) What is its mixing time? *Hint*: The mixing time does not depend on n .
- (d) Now consider a new Markov chain with state space Ω and transition probabilities $\hat{P}_{i,j} = P_{j,i}\pi_j/\pi_i$. (This is the time-reversal of the original chain.) Find the stationary distribution of the new chain, and show that, wherever it starts, it is in its stationary distribution after n steps (so the mixing time is at most n).